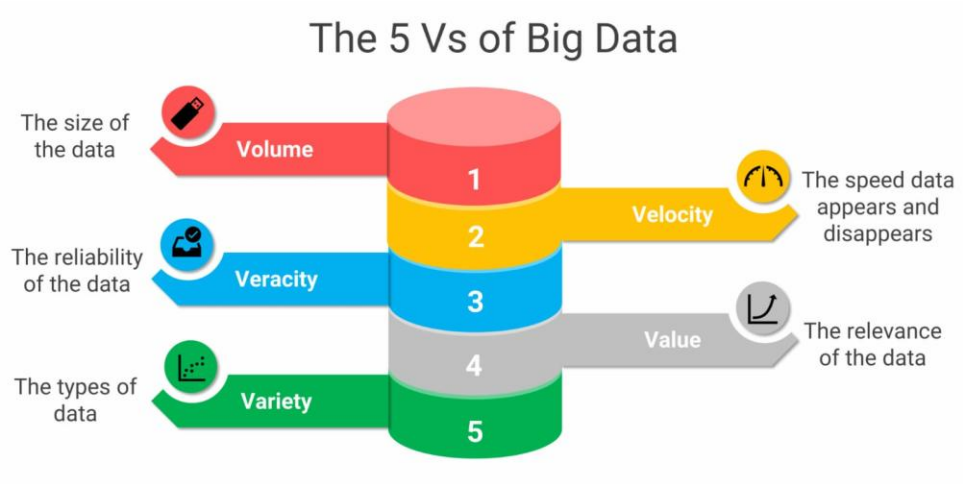
Big Data Process Mapping – Amazon Recommendation System

1.INTRODUCTION:

Big Data refers to extremely large and complex data sets that cannot be processed using traditional data processing tools. In today's digital world, huge amounts of data are generated every second from websites, mobile applications, online shopping platforms, and social media.

Big Data is mainly defined by 5 V's:

- **Volume** – Large amount of data
- **Velocity** – Speed at which data is generated
- **Variety** – Different types of data (text, images, videos)
- **Veracity** – Accuracy and reliability of data
- **Value** – Useful insights obtained from data



Many companies use big data to improve their services. One of the best real-world examples is Amazon. Amazon uses big data analytics to recommend products to customers based on their interests and behavior.

This report explains the complete big data process mapping of Amazon's recommendation system including data sources, storage, processing, and output.

Big data is used in many fields such as:

- Healthcare
- Banking
- Education
- E-commerce

- Smart cities

One of the best examples of big data usage is Amazon recommendation system. It helps customers find products easily and improves user experience.

2.OVERVIEW OF AMAZON RECOMMENDATION SYSTEM

The Amazon recommendation system is one of the most advanced real-world applications of big data and machine learning. It is designed to analyze customer behavior and provide personalized product suggestions to users. This system helps customers easily find products they may like and also helps Amazon increase its sales and customer satisfaction.

Amazon is one of the world's largest e-commerce platforms with millions of users visiting the website and mobile app every day. Each user generates a large amount of data through their activities such as searching for products, clicking on items, adding products to the cart, purchasing items, and giving reviews. All this information is collected and analyzed by Amazon's big data system.

The recommendation system studies customer behavior and provides suggestions based on individual interests. For example, when a user searches for a mobile phone, Amazon shows similar mobile phones, accessories, and related products. When a customer buys a product, the system suggests other products that are frequently bought together.

2.1 How Amazon Recommendation System Works

The system works by tracking and analyzing user activities on the Amazon platform. Whenever a customer performs any action such as searching, clicking, or purchasing, the system records that information. This collected data is then processed using machine learning algorithms.

The recommendation system compares the behavior of one user with millions of other users. If many users who bought a particular product also bought another product, the system will recommend that product to new customers. This helps in providing accurate and useful suggestions.

2.2 Types of Recommendations Provided by Amazon

Amazon provides different types of product recommendations such as:

1. **Personalized Recommendations**
   Based on user search history and previous purchases.
2. **Frequently Bought Together**
   Shows products that are commonly purchased together.
3. **Customers Who Bought This Also Bought**
   Suggests products purchased by similar users.

4. **Trending Products**
   Shows popular and trending items.
5. **Recently Viewed Items**
   Displays products that the user has viewed recently.

These recommendations help customers discover new products easily and save time while shopping.

2.3 Objectives of Amazon Recommendation System

The main objectives of Amazon's recommendation system are:

- To provide personalized shopping experience
- To understand customer interests and behavior
- To increase sales and profit
- To improve customer satisfaction
- To provide relevant product suggestions
- To enhance marketing strategies

By using big data analytics, Amazon can understand what customers want and deliver better services. This system also helps Amazon maintain strong customer relationships.

3. Data Sources in Amazon Recommendation System

Data collection is the first and most important step in the big data process mapping of the Amazon recommendation system. Amazon collects a huge amount of data from different sources every second. This data helps the system understand customer behavior and provide personalized recommendations.

Amazon collects both **structured data** and **unstructured data**. Structured data includes numbers and text such as price and quantity, while unstructured data includes images, videos, and reviews.

3.1 User Data

User data is the most important data source for Amazon. Every activity performed by a user on the Amazon website or mobile app is recorded.

User data includes:

- Search history
- Browsing history
- Time spent on a product
- Click patterns
- Wishlist items

- Items added to cart
- Recently viewed products

For example, if a user searches for shoes and views sports shoes multiple times, the system understands that the user is interested in sports shoes and recommends similar products.

This helps Amazon understand customer interests and preferences.

3.2 Product Data

Product data includes all information related to products available on Amazon.

Product data includes:

- Product name
- Brand
- Category
- Price
- Description
- Images
- Ratings and reviews
- Stock availability

This data helps the recommendation system match customer interests with suitable products. For example, if a user searches for a budget mobile phone, the system recommends mobiles in that price range.

3.3 Transaction Data

Transaction data is generated when a customer makes a purchase. This is very important for recommendation systems.

Transaction data includes:

- Payment details
- Order history
- Delivery address
- Order date and time
- Return and refund details

By analyzing transaction data, Amazon understands what products customers buy frequently and which products are often purchased together. This helps in suggesting related products.

3.4 Review and Rating Data

Customers give ratings and reviews after purchasing products. This is also an important data source.

It includes:

- Star ratings
- Written reviews
- Feedback
- Product comments

Positive reviews increase product recommendations, while negative reviews may reduce suggestions. This helps Amazon maintain quality and customer satisfaction.

3.5 Device and Location Data

Amazon also collects technical data such as:

- Device type (mobile, laptop, tablet)
- Location and region
- Language preferences
- Time of usage

This helps Amazon recommend products based on location and trends. For example, winter clothes may be recommended in cold regions.

3.6 External Data Sources

Amazon also uses external data such as:

- Market trends
- Festival season data
- Social media trends
- Customer feedback

For example, during festivals like Diwali, Amazon recommends gifts, decorations, and offers.

4. Data Storage

After collecting data, it must be stored securely and efficiently.

Amazon uses advanced cloud-based storage systems such as:

- **Data warehouses**
- **Data lakes**
- **Cloud storage systems**

4.1 Data Lake

A data lake stores raw data in its original format.
It can store:

- Text
- Images
- Videos
- Logs

4.2 Data Warehouse

A data warehouse stores processed and structured data.
It is used for:

- Reporting
- Analysis
- Business intelligence

4.3 Distributed Storage

Since data volume is very large, Amazon uses distributed storage systems where data is stored across multiple servers. This ensures:

- High speed
- Reliability
- Backup and recovery

5. Data Processing

Data processing is the most important stage in big data systems.

Amazon uses big data processing frameworks like:

- Hadoop
- Spark
- Machine Learning algorithms

5.1 Data Cleaning

Raw data may contain errors or duplicate entries.
Data cleaning involves:

- Removing duplicates
- Correcting errors
- Filling missing values

5.2 Data Transformation

Data is converted into structured format for analysis:

- Organizing data into tables
- Converting formats
- Categorizing products

5.3 Data Analysis

Machine learning algorithms analyze user behavior:

- What users search
- What they buy
- How long they view products
- Which products they ignore

5.4 Recommendation Algorithms

Amazon uses algorithms such as:

- Collaborative filtering
- Content-based filtering
- Hybrid recommendation systems

These algorithms predict what a user may like based on past behavior and similar users.

6. Data Output

After processing, the system generates useful outputs.

6.1 Personalized Recommendations

Users see:

- Recommended products
- Similar items
- Best deals
- Trending products

6.2 Business Insights

Amazon gets insights like:

- Customer preferences
- Popular products
- Sales trends
- Market demand

6.3 Notifications

- Email suggestions
- App notifications
- Special offers

These outputs help improve customer satisfaction and increase company profit.